Literature Review Outline & Bibliography
Layer-wise Attention Entropy for Real-Time Hallucination Detection in Small LLMs: A Comparative Study Across General, Medical, and Legal Domains
Divyavaahini Thyagarajan · Venkatesh Lankalapalli · Sai Sreenivas Putta
EPITA School of Engineering | Action Learning Program, Spring 2026 | May 29, 2026

---

1. Problem Context

Large Language Models (LLMs) generate outputs that are fluent, confident, and occasionally factually incorrect, a phenomena known as hallucinations. The issue is architectural: LLMs produce the next token based on learnt statistical patterns rather than reviewing confirmed data. When a pattern is ambiguous or absent, the model nonetheless generates a response that may not be supported by reality (Rawte et al., 2024).

A special and unexplored dimension addresses small models with 1 to 3 billion parameters. These models are commonly employed in resource constrained situations, although practically all hallucination detection research has concentrated on 7B+ parameter models (Chuang et al., 2024a; Chen et al., 2024; Binkowski et al., 2025). It is unclear whether detection signals apply to small models.

A further unresolved challenge is cross-domain transfer. A detector trained on general QA data may not perform reliably on medical or legal language. Pandit et al. (2025) demonstrated that even GPT-4o achieves an F1 as low as 0.625 on the hardest medical hallucination cases. Hou et al. (2024) found that 80% of LLM-generated legal analyses contain hallucinations. No study has examined whether attention-based detectors trained on general QA transfer to these specialist domains at 1B-3B scale.

Why Is It Important?

In clinical decision support, a hallucinated drug interaction or diagnostic advice can endanger patients. In legal practice, a falsified case citation might lead to confusion. In teaching, confidently incorrect responses spread misinformation. These are precisely the conditions in which small, efficient LLMs are used, and where existing detection research provides the least coverage. This research bridges the gap between three model designs, three assessment domains, and three families of detection algorithms.

2. Overview of Existing Work & Comparison of Approaches

Sahoo et al. (2024) organise the hallucination detection literature into three method families. This review follows that taxonomy, as it maps directly to the three experimental tasks this project implements.

Family	Key Papers	Strengths	Weaknesses
Consistency-based	Su et al. (2024)	No model internals; no annotation; real-time via MIND	Multiple forward passes (K=5-10); high latency; not validated on sub-3B models

Family	Key Papers	Strengths	Weaknesses
Output-entropy	Farquhar et al. (2024); Nikitin et al. (2024)	Statistically grounded; proven; strong results on large models	Requires K sampled responses; fails on confident hallucinations; slow
Attention / internal-state	Chuang et al. (2024a); Chen et al. (2024); Binkowski et al. (2025); Han et al. (2025)	Single forward pass; interpretable; current SOTA for attention-based detection	All prior work on 7B+ models only; cross-domain transfer not studied

The key distinction for this research is that only internal-state algorithms are capable of true real-time detection. Janiak et al. (2025) express worry that detection findings are consistently inflated when analyzed using ROUGE methods, which can be reduced by up to 45.9% when evaluated correctly. Kulkarni et al. (2025) emphasize the importance of using various complementing criteria to ensure a reliable evaluation. Furthermore, Chuang et al. (2024b) shown in DoLa that factual knowledge in transformer models is not distributed uniformly across levels, but is concentrated in specific middle-to-upper layers. This offers the theoretical basis for layer-wise analysis.

Attention interpretability debate

A fundamental and unresolved question is whether attention weights are truly interpretable or simply associated artifacts. Attention weights are the result of a SoftMax oversampled dot product and do not exactly correspond to information flow in the network. Instead of factual doubt, high attention entropy could indicate grammatical complexity, token rarity, or sequence length. Binkowski et al. (2025) recognize this and regard attention mappings as graph-structured signals rather than interpretable weights. This initiative does not suggest that attention entropy causes hallucinations; rather, it treats entropy as a valid correlation signal suitable for deployment.

Confident hallucination failure

The most harmful hallucinations occur when the model generates an incorrect answer with high confidence and low entropy, which is routinely ignored by uncertainty-based detectors. Janiak et al. (2025) show that these circumstances are the principal failure mode of all entropy-based approaches. Kulkarni et al. (2025) demonstrate that no one metric reliably captures them, which directly supports this project's four-metric evaluation protocol: AUROC, F1, ECE, and latency.

Shannon vs semantic vs spectral comparison

	Shannon Entropy (Task 1)	**Semantic Entropy (Farquhar et al., 2024)**	**Spectral Attention (Binkowski et al., 2025)**
Passes needed	1 forward pass	K=5 sampled responses	1 forward pass
Signal used	Internal attention weights	Generated text meaning clusters	Attention graph structure

	Shannon Entropy (Task 1)	**Semantic Entropy (Farquhar et al., 2024)**	**Spectral Attention (Binkowski et al., 2025)**
Speed	Fast real-time capable	5× slower	Moderate
Fails when	Model confidently wrong	All responses wrong	Architecturally unstable
Model size tested	1B-3B (this project)	7B+ only	7B+ only

3. Identified Gap & Project Positioning

Gap 1: The small-model regime is uncharted. Every attention-based and internal-state detection paper (Lookback Lens, INSIDE, Spectral Attention Maps, Simple Factuality Probes) relied solely on 7B+ models. It is unclear if these signals exist, endure, or decay at the 1B-3B scale.

Gap 2: There is no layer-wise entropy investigation. Neither Chuang et al. (2024a) nor Binkowski et al. (2025) investigated whether transformer layer contains the greatest discriminative entropy signal. DoLa demonstrates that factual knowledge is layer-localised, but no one has turned this into a layer-specific detection strategy.

Gap 3: Cross-domain transfer at the tiny scale has not been studied. No work has investigated whether a probe trained on general QA can transfer zero-shot to medical and legal domains on 1B-3B models, or measured the performance deterioration.

Gap 4: The calibration reliability of attention-based probes has not been examined. No work has investigated whether attention entropy probe confidence ratings are well-calibrated for tiny models. Kulkarni et al. (2025) show that detectors are typically overconfident, even when AUROC appears robust. For safe deployment in medical and legal situations, ECE is just as crucial as AUROC; an overconfident detector is more damaging than a conservative one.

This project fills all four gaps simultaneously by studying layer-wise Shannon attention entropy as a hallucination detection signal profiling entropy at every transformer layer on Llama-3.2-1B, Llama-3.2-3B, and Qwen2.5-1.5B the first study at this scale. Three independent tasks address one method family each, sharing identical models, datasets, and evaluation protocols (AUROC, F1, ECE, latency) per the recommendations of Janiak et al. (2025) and Kulkarni et al. (2025).

Correlation vs causation

We acknowledge that attention entropy is a correlation signal, not a causal explanation the claim is that entropy reliably co-occurs with hallucination at specific layers in 1B-3B models, which is sufficient for real-time deployment.

4. Research-to-Application Mapping

Research Insight	Source	How We Apply It
Attention ratios detect hallucinations in one forward pass	Chuang et al. (2024a)	Extend to full layer-wise Shannon entropy sweep on 1B-3B models
Factual knowledge concentrated in middle-to-upper transformer layers	Chuang et al. (2024b)	Profile entropy per layer; check alignment with DoLa's factual-injection layers
Spectral attention features are current SOTA	Binkowski et al. (2025)	Compare Shannon entropy vs. Laplacian eigenvalue features on same test sets
Output entropy over meaning clusters beats token-level perplexity	Farquhar et al. (2024)	Implement semantic entropy as probability baseline; compare against attention entropy
Unsupervised real-time detection achievable from internal states	Su et al. (2024)	Adopt their real-time objective and latency protocol to validate 15% overhead constraint
Detection results overstated when evaluated with ROUGE	Janiak et al. (2025)	Use LLM-as-Judge evaluation; report AUROC, F1, ECE, and latency
Medical and legal hallucination unsolved even for SOTA models	Pandit et al. (2025); Hou et al. (2024)	Test zero-shot cross-domain transfer to both domains to quantify performance degradation

5. Annotated Bibliography - 15 Validated Sources

All sources are peer-reviewed, published 2024-2025, and directly aligned with the project's three experimental tasks and evaluation design.

#	Authors	Title	Venue/Year	Type	Cluster
1	Chuang et al.	Lookback Lens: Detecting and Mitigating Contextual Hallucinations Using Only Attention Maps	EMNLP 2024	Core	Attention

#	Authors	Title	Venue/Year	Type	Cluster
2	Chen et al.	INSIDE: LLMs' Internal States Retain the Power of Hallucination Detection	ICLR 2024	Core	Internal-state
3	Han et al.	Simple Factuality Probes Detect Hallucinations in Long-Form NLG	EMNLP Findings 2025	Core	Probe
4	Farquhar et al.	Detecting Hallucinations in LLMs Using Semantic Entropy	Nature 2024	Core	Probability
5	Su et al.	Unsupervised Real-Time Hallucination Detection Based on Internal States of LLMs	ACL Findings 2024	Core	Consistency
6	Binkowski et al.	Hallucination Detection in LLMs Using Spectral Features of Attention Maps	EMNLP 2025	Core	Attention
7	Chuang et al.	DoLa: Decoding by Contrasting Layers Improves Factuality in LLMs	ICLR 2024	Core	Layer-wise
8	Bang et al.	HalluLens: LLM Hallucination Benchmark	ACL 2025	Core	Benchmark
9	Sahoo et al.	A Comprehensive Survey of Hallucination in Foundation Models	EMNLP Findings 2024	Core	Survey
10	Nikitin & Marttinen	Kernel Language Entropy: Fine-grained Uncertainty Quantification for LLMs	NeurIPS 2024	Core	Probability
11	Janiak et al.	The Illusion of Progress: Re-evaluating Hallucination Detection in LLMs	EMNLP 2025	Supporting	Evaluation
12	Kulkarni et al.	Evaluating Evaluation Metrics: The Mirage of Hallucination Detection	EMNLP Findings 2025	Supporting	Evaluation
13	Pandit et al.	MedHallu: A Comprehensive Benchmark for Detecting Medical Hallucinations in LLMs	EMNLP 2025	Supporting	Medical

#	Authors	Title	Venue/Year	Type	Cluster
14	Hou et al.	Gaps or Hallucinations? Scrutinizing Machine-Generated Legal Analysis	NLLP @ EMNLP 2024	Supporting	Legal
15	Rawte et al.	Tutorial Proposal: Hallucination in Large Language Models	LREC-COLING 2024	Supporting	Background

6. Full Reference List (APA 7th Edition)

Bang, Y., Ji, S., Schelten, A., Hartshorn, P., Fowler, T., Zhang, L., Cancedda, N., & Fung, P. (2025). HalluLens: LLM hallucination benchmark. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics. https://aclanthology.org/2025.acl-long.1176/

Binkowski, M., Janiak, M., Sawczyn, A., Gabrys, B., & Kajdanowicz, T. (2025). Hallucination detection in LLMs using spectral features of attention maps. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics. https://aclanthology.org/2025.emnlp-main.1239/

Chen, H., Liu, R., Chen, D., Gu, Q., Wu, Y., Tao, C., Fu, J., & Ye, J. (2024). INSIDE: LLMs' internal states retain the power of hallucination detection. In *Proceedings of the 12th International Conference on Learning Representations*. OpenReview.net. https://proceedings.iclr.cc/paper_files/paper/2024/hash/0d1986a61e30e5fa408c81216a616e20-Abstract-Conference.html

Chuang, Y., Qiu, L., Hsieh, C., Krishna, R., Kim, Y., & Glass, J. (2024a). Lookback lens: Detecting and mitigating contextual hallucinations in large language models using only attention maps. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics. https://aclanthology.org/2024.emnlp-main.84/

Chuang, Y., Xie, Y., Luo, H., Kim, Y., Glass, J., & He, P. (2024b). DoLa: Decoding by contrasting layers improves factuality in large language models. In *Proceedings of the 12th International Conference on Learning Representations*. OpenReview.net. https://proceedings.iclr.cc/paper_files/paper/2024/hash/edc36117f795ca52a0cbf6a7b3882859-Abstract-Conference.html

Farquhar, S., Kossen, J., Kuhn, L., & Gal, Y. (2024). Detecting hallucinations in large language models using semantic entropy. *Nature*, *630*, 625-630. https://doi.org/10.1038/s41586-024-07421-0

Han, S., Band, N., Razzak, S., Kossen, J., Rudner, T. G. J., & Gal, Y. (2025). Simple factuality probes detect hallucinations in long-form natural language generation. In *Findings of the Association for Computational Linguistics: EMNLP 2025*. Association for Computational Linguistics. https://aclanthology.org/2025.findings-emnlp.880/

Hou, Y., Jurayj, J., Holzenberger, N., Blair-Stanek, A., & Van Durme, B. (2024). Gaps or hallucinations? Scrutinizing machine-generated legal analysis for fine-grained text evaluations. In

Proceedings of the Natural Legal Language Processing Workshop 2024. Association for Computational Linguistics. https://aclanthology.org/2024.nllp-1.24/

Janiak, M., Binkowski, M., Sawczyn, A., Gabrys, B., Shwartz-Ziv, R., & Kajdanowicz, T. (2025). The illusion of progress: Re-evaluating hallucination detection in LLMs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics. https://aclanthology.org/2025.emnlp-main.1761/

Kulkarni, A., Zhang, X., Moniz, J. R. A., Ge, T., Tseng, B., Piraviperumal, D., Swayamdipta, S., & Yu, H. (2025). Evaluating evaluation metrics: The mirage of hallucination detection. In *Findings of the Association for Computational Linguistics: EMNLP 2025*. Association for Computational Linguistics. https://aclanthology.org/2025.findings-emnlp.1035/

Nikitin, A., & Marttinen, P. (2024). Kernel language entropy: Fine-grained uncertainty quantification for LLMs from semantic similarities. In *Proceedings of the 38th Annual Conference on Neural Information Processing Systems*. Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2024/hash/10c456d2160517581a234dfde15a7505-Abstract-Conference.html

Pandit, A., Xu, M., Hong, P., Wang, J., Chen, M., Xu, Y., & Ding, J. (2025). MedHallu: A comprehensive benchmark for detecting medical hallucinations in large language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics. https://aclanthology.org/2025.emnlp-main.143/

Rawte, V., Chadha, A., Sheth, A., & Das, A. (2024). Tutorial proposal: Hallucination in large language models. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation*. ELRA and ICCL. https://aclanthology.org/2024.lrec-tutorials.11/

Sahoo, P., Meharia, P., Ghosh, A., Saha, S. K., Jain, V., & Chadha, A. (2024). A comprehensive survey of hallucination in large language, image, video and audio foundation models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*. Association for Computational Linguistics. https://aclanthology.org/2024.findings-emnlp.685/

Su, H., Wang, H., Ai, B., Hu, X., Wu, Z., Zhou, L., & Liu, Y. (2024). Unsupervised real-time hallucination detection based on the internal states of large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*. Association for Computational Linguistics. https://aclanthology.org/2024.findings-acl.854/

Limitations :

Two genuine constraints influenced the project design. First, entropy cannot detect confident hallucinations by design; if the model is incorrect with low entropy, the probe will miss them. Second, because layer instability varies across architectures, the detection-optimal layer number differs between Llama-1B (16 layers) and Qwen/Llama-3B (28 layers). To enable meaningful cross-architecture comparison, results are compared by relative depth (layer index divided by total layers), rather than absolute layer number.

7. Conclusion

The review of 15 peer-reviewed papers from 2024-2025 yielded three principal findings. First, internal-state detection methods particularly those using attention distributions are the most promising direction for efficient hallucination detection, operating in a single forward pass. Second, the 1B-3B model scale is a genuine research gap: all published attention-based work has evaluated exclusively on 7B+ parameter models. Third, no paper has conducted a layer-by-layer analysis of attention entropy as a hallucination signal, despite DoLa (Chuang et al., 2024b) providing the theoretical basis for expecting this analysis to be informative.

Reading the books directly influenced the project design. DoLa motivated layer-by-layer entropy profiling; Janiak et al. (2025) and Kulkarni et al. (2025) drove the decision to use four complementary evaluation metrics; Pandit et al. (2025) and Hou et al. (2024) confirmed medical and legal domains as the most demanding and practically consequential targets for cross-domain evaluation; and Binkowski et al. (2025) established the spectral attention method as the direct state-of-the-art comparison point for Shannon entropy featur

Flesch Reading Ease Score :